

Name: Key

CHE 321; Spring - 2020; Quiz - 1

100 Points

1. (25 points) What is the meaning of 'order of reaction'?

Order of reaction refer to power to which the concentration is raised

$$r = k C_A^\alpha C_B^\beta$$

This reaction is α order w.r.t A ; β - order w.r.t B .

2. (25 points) For a single liquid-phase reaction, which reactor will lead to a higher conversion: A plug-flow reactor (PFR) or a continuous stirred tank reactor (CSTRs)?

+25 (PFR)

CSTR

3. (25 points) A catalytic CSTR reactor of volume V (m^3) has W (Mass) Kgs of the catalyst. What is the relationship between k_A and k'_A for this reactor? Note, the bulk density of the catalyst is $\rho_B = W/V$.

$$V = \frac{F_{j0} - F_j}{-r_j} ; W = \frac{F_{j0} - F_j}{-r'_j} + 15$$

PFR: -5.

$$\Rightarrow V r_j = W r'_j = \rho_B V r'_j \Rightarrow r_j = \rho_B r'_j + 5$$

$$\Rightarrow \boxed{k_A = \rho_B k'_A} + 5$$

4. (25 points) Qualitatively, draw a Levenspiel's plot for two CSTRs in series.

